

Elimination of the GGHV Branch-(c) Normal Form at Degree Pair (72, 108):

A Certified Computational Audit
Descent, Chart Conditions, and the Rank-Lemma Obstruction

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Computation, Lean and LaTeX Source:

[https://github.com/brandonmccraryresearch-cloud/
2D-Jacobian-Conjecture-Workspace](https://github.com/brandonmccraryresearch-cloud/2D-Jacobian-Conjecture-Workspace)

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Abstract

We present a computational elimination of the branch-(c) normal form arising from case (1) of Proposition 4.3 of Guccione–Guccione–Horruitiner–Valqui [1] for a hypothetical (72, 108) counterexample to the planar Jacobian conjecture. Every base object is source-gated directly against GGHV v1 before any proof is drafted. This paper is the branch-(c) companion to the branch-(a,b) elimination [2], whose format it follows.

Starting from the verified Newton polygons

$$\begin{aligned} N(P) &= \text{conv}\{(0, 0), (1, 0), (8, 14), (8, 16), (0, 8)\}, \\ N(Q) &= \text{conv}\{(0, 0), (2, 1), (12, 21), (12, 24), (0, 12)\}, \end{aligned}$$

with $[P, Q] = \lambda x^2$ post- φ and $\lambda \neq 0$, we reduce the bracket condition to layer identities E_n in the grading $w(x) = 2, w(y) = -1$, extract 132 exact scalar equations, and eliminate the layers $E_4, \dots, E_{-2}$ to twelve chart conditions $\Omega, \Psi, \Phi_1, \Phi_2, \Theta_1, \Theta_2, \Theta_3$ plus five pure rows, over the quintic field $K_5 = \mathbb{Q}[w]/(w^5 - w^4 + 3w^3 + 3w^2 + 26)$. The top layer is the branch-(a,b) problem, whose exact K_5 classification is inherited. The descent is proved twice: symbolically, as 600 kernel-checked reflective polynomial identities in Lean 4 (no sorry, standard axioms only), and numerically, by exact finite-field linear algebra.

The chart is empty. On the stratum $b_{11,20} = 0$ this is proved in the Lean kernel from an exact K_5 certificate (degree 2, 75 terms). On the whole chart it follows from a rank lemma: full row rank of the fixed-degree Macaulay matrix lifts from $\mathbb{F}_p$ to K_5 through a nonzero minor. At weight $W = 24$ the 3199×6054 Macaulay matrix has full row rank 3199 at two primes ($p = 1000003, p = 32003$), verified by two independent implementations (FLINT and NumPy), with negative controls. An earlier reduction lemma used in a preliminary version of this work is false as stated and is retracted here; the rank lemma is its correct replacement, and the distinction—rank lifts, mere consistency does not—is proved by counterexample.

In Lean, with all 790 modules built and `#print axioms` reporting only `propext`, `Classical.choice`, `Quot.sound`,

$$\begin{aligned} \text{ChartEmptyC_T1ne0 L} &\Rightarrow \neg \exists P Q \lambda, \lambda \neq 0 \wedge \text{NewtonNFC } P Q \\ &\wedge [P, Q] = \lambda x^2, \end{aligned}$$

and the rank computation supplies `ChartEmptyC\_T1ne0` outside Lean, at epistemic grade B. The application to the Jacobian conjecture is conditional on the correctness and exhaustiveness of [1, Proposition 4.3] (Corollary 1.4).

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1 Introduction

1.1 Context and prior work

The planar Jacobian conjecture asserts that a polynomial map $(P, Q) : \mathbb{C}^2 \rightarrow \mathbb{C}^2$ with $[P, Q] := P_x Q_y - P_y Q_x \in \mathbb{C}^\times$ is invertible. In [1], Guccione, Guccione, Horruitiner, and Valqui raise the lower bound on $\max\{\deg P, \deg Q\}$ for a hypothetical counterexample from 100 (Moh [5]) to 108, leaving open only the degree pair (72, 108) and its symmetric (108, 72).

For the degree pair (72, 108), [1] (an unrefereed preprint, arXiv:2204.14178v1) identifies two cases. The corner (9, 27) is discarded in [1, §5]; the open case is the corner (8, 28) with $(m, n) = (3, 2)$. For this case, [1, Proposition 4.3] shows that any counterexample gives rise, after an explicit (non-invertible) morphism φ , to one of two Newton polygon configurations. Case (2), branches (a) and (b), is eliminated in [2]. Case (1), the subject of this paper, is branch (c):

$$\begin{aligned} N(P) &= \text{conv}\{(0, 0), (1, 0), (8, 14), (8, 16), (0, 8)\}, \\ N(Q) &= \text{conv}\{(0, 0), (2, 1), (12, 21), (12, 24), (0, 12)\}. \end{aligned} \tag{1}$$

with $[P, Q] = \lambda x^2$ in post- φ coordinates and $\lambda \neq 0$. The additional vertices (0, 8) and (0, 12) distinguish branch (c) from branches (a,b). See Figure 1 for the polygon shapes.

FIGURE PLACEHOLDER 1

What goes here (Figure 1). Two side-by-side Newton polygon plots, rendered from the exact vertex data of (1), source-gated at GGHV v1 lines 495 and 599–600. *Left panel:* $N(P) = \text{conv}\{(0, 0), (1, 0), (8, 14), (8, 16), (0, 8)\}$ with all lattice points marked (25 from the branch-(a,b) sub-polygon plus the points contributed by the (0, 8) vertex). *Right panel:* $N(Q) = \text{conv}\{(0, 0), (2, 1), (12, 21), (12, 24), (0, 12)\}$ with all lattice points marked. In both panels the branch-(a,b) sub-polygon $\text{conv}\{(0, 0), (1, 0), (8, 14), (8, 16)\}$ resp. $\text{conv}\{(0, 0), (2, 1), (12, 21), (12, 24)\}$ is drawn in a lighter shade underneath, and the extra branch-(c) vertices (0, 8) and (0, 12) are highlighted (e.g. in the accent color), with an arrow and label indicating “branch-(c) extension”. The ten vertex coefficients required nonzero by `NewtonNFC` are labelled at the vertices. Rendering: vector PDF (not raster), produced by a short Python script from the vertex lists, so the figure is regenerable from source; axes labelled (i, j) for the monomial $x^i y^j$.

Santibáñez-Leal has published machine evidence related to the (72, 108) candidate [6, 7], explicitly not claiming elimination. The candidate itself is documented in [8]. Independent non-peer-reviewed treatments of the GGHV configuration in 2026 include Helali’s exact characteristic-zero certificate [9] and the structural analysis of wstrinz [10]; none is peer-reviewed, and the parity-constrained Davenport–Stothers equation relevant to the top layer remains unclassified.

1.2 Main result

Definition 1.1 (Branch-(c) Newton normal form). Let L be a field of characteristic 0 and $P, Q \in L[x, y]$. `NewtonNFC`(P, Q) holds when $\text{supp } P \subset N(P)$ and $\text{supp } Q \subset N(Q)$ with $N(P), N(Q)$ as in (1), and the ten vertex coefficients (at (0, 0), (1, 0), (8, 14), (8, 16), (0, 8) of P

and $(0, 0), (2, 1), (12, 21), (12, 24), (0, 12)$ of Q) are all nonzero. In Lean this is `NewtonNFc` (`Jacobian/BranchC/DescentClaim.lean`); see listing 1.

Theorem 1.2 (Elimination of the GGHV branch-(c) normal form, conditional). *Let L be a field of characteristic 0 satisfying `ChartEmptyC_T1ne0` (listing 3): the twelve chart conditions of §8 have no common zero with $b_{12,22} = 1$ on the stratum $b_{11,20} \neq 0$. Then there are no $P, Q \in L[x, y]$ and no $\lambda \neq 0$ with `NewtonNFc`(P, Q) and $[P, Q] = \lambda x^2$.*

The hypothesis `ChartEmptyC_T1ne0` is a precise Lean Prop. It is established outside Lean in §10, at epistemic grade B (see §12).

Corollary 1.3 (Unconditional elimination under computational premises). *Under the computational premises of §10 (the rank lemma, Lemma 10.1, instantiated at two primes by two independent implementations, with negative controls), `ChartEmptyC_T1ne0` holds for every field L of characteristic 0, and hence no P, Q, λ as in Theorem 1.2 exist.*

Corollary 1.4 (Jacobian application). *Assume [1, Proposition 4.3] is correct and exhaustive. Then no counterexample to the planar Jacobian conjecture of degree pair $(72, 108)$ arises from branch (c).*

1.3 Logical status

Remark 1.5. Theorem 1.2 does not depend on GGHV: its Lean proof goes from `NewtonNFc` and $[P, Q] = \lambda x^2$ through the descent to the chart statement, all inside Lean. In Lean, `main_theorem_c_of_chartEmpty_T1ne0` (listing 4) derives the conclusion from the explicit hypothesis `ChartEmptyC_T1ne0` (listing 3), with `#print axioms` reporting only `propext`, `Classical.choice` and `Quot.sound`, and no `sorry`, `axiom` or `native_decide` anywhere in the 790 modules. The chain is: `NewtonNFc + [P, Q] = λx²` $\xrightarrow{B0-B3}$ `Descent-ClaimC` hypotheses $\xrightarrow{B4-B5}$ `ChartEmptyC` $\xrightarrow{B6-B7}$ `ChartEmptyC_T1ne0` $\Rightarrow$ contradiction. Only Corollary 1.4 additionally assumes GGHV Proposition 4.3.

Remark 1.6 (A retraction). A preliminary version of this work claimed a “reduction lemma”: that a unit ideal modulo a prime lifts to a unit ideal over $\mathbb{Q}$. That statement is **false**. Counterexample: $F = 1 + 109X$ is 1 modulo 109 (unit ideal) but vanishes at $X = -1/109$ over $\mathbb{Q}$ (proper ideal). The Lean file deriving `False` from the stated axiom is archived for the record, and every theorem depending on it is vacuous. The correct replacement is the rank lemma (Lemma 10.1): *full row rank* of a fixed-degree Macaulay matrix lifts through a nonzero minor, because nonvanishing of a determinant is an open condition. Mere consistency of a modular system does not lift. §12 gives the precise distinction.

1.4 Outline

Section 2 source-gates all base objects against GGHV v1. Section 3 sets up the polynomial system and the grading. Section 4 gives the exact layer reduction and the 132 raw equations. Section 5 fixes normalizations and reaches the chart. Section 6 treats the top layer, inherited from branches (a,b). Section 7 carries out the descent to the twelve conditions, proved by kernel reflection. Section 8 studies the chart conditions. Section 9 eliminates the stratum $b_{11,20} = 0$ inside Lean. Section 10 eliminates the whole chart by the rank lemma. Section 11 assembles the proof of Theorem 1.2. Section 12 gives the epistemic accounting: grades, the retraction, and what is not claimed. Section 13 concludes. Section 14 covers prior work, attribution, and independence. Figure 2 shows the bridge architecture.

FIGURE PLACEHOLDER 2

What goes here (Figure 2). A flowchart of the nine bridges B0–B8 (Table 11), laid out left to right in three bands. *Top band (Lean, grade A)*: boxes for B0 (layer identities), B1 (top layer), B2 (degree-19 rigidity, $t_2 \neq 0$), B3 (normalization to chart), B4 (coefficient extraction, 132 equations), B5 (eliminations $E_4 \dots E_{-2}$, 600 reflective identities), B6 (split on $b_{11,20}$), B7 (stratum $b_{11,20} = 0$ in the kernel). *Middle band (outside Lean, grade B)*: box for B8 (rank lemma at $W = 24$, two primes). *Bottom band*: the Lean propositions each bridge targets (EIdent, DescentClaimC, ChartEmptyC, ChartEmptyC_T1ne0, main_theorem_c_of_chartEmpty_T1ne0), with arrows showing composition to the final contradiction. Each box lists its Lean file/theorem name and its epistemic grade. Rendering: vector PDF (TikZ or Graphviz dot), regenerable from a checked-in source file; box colors encode grade (A/B) consistently with Table 12.

2 Source data

Every base object below is source-gated directly against GGHV v1 before any proof is drafted, following the discipline of [2]. The branch-(c) Newton polygons (1) are [1, Proposition 4.3, case (1)]: the vertex lists are asserted at GGHV v1 lines 495 and 599–600, and the checker `source_gate_verify_c.py` re-asserts them byte-for-byte against the downloaded v1 text. The post- ϕ bracket target $[P, Q] = \lambda x^2$ is [1, §5].

Listing 1: The branch-(c) Newton normal form (NewtonNFc)

```
-- Jacobian/BranchC/DescentClaim.lean (excerpt, exact statement)
def NewtonNFc (P Q : MvPolynomial (Fin 2) L) : Prop :=
  (P.support \subseteq NPc) \land (Q.support \subseteq NQc) \land
  TenVertexCoeffsNonzero P Q
-- NPc = conv {(0,0),(1,0),(8,14),(8,16),(0,8)}
-- NQc = conv {(0,0),(2,1),(12,21),(12,24),(0,12)}
```

The Lean formalization works over an arbitrary field L of characteristic 0. The quintic field

$$K_5 = \mathbb{Q}[w]/(R), \quad R(w) = w^5 - w^4 + 3w^3 + 3w^2 + 26, \quad (2)$$

enters through the top layer (§6): R is irreducible over $\mathbb{Q}$, and it remains irreducible modulo 109, so $\mathbb{F}_{109^5} = \mathbb{F}_{109}[w]/(R)$ is a field. The good prime/root pairs used in this audit are

$$(101, 9), \quad (1000003, 806739), \quad (32003, 11147), \quad (3)$$

each satisfying $R(w_0) \equiv 0 \pmod{p}$, plus the inert prime 109.

3 The polynomial system

3.1 Lattice points

Write $P = \sum a_{i,j} x^i y^j$, $Q = \sum b_{i,j} x^i y^j$. The branch-(c) supports are the lattice points of the polygons (1); the supports strictly contain those of branches (a,b) because of the vertices (0, 8) and (0, 12). The Lean files record the supports as `rangeP`, `rangeQ` (Jacobian/BranchC/DescentClaim.lean).

3.2 The bracket equations

Grade by $w(x) = 2$, $w(y) = -1$. Write the layers

$$P = \sum_k y^{-k} A_k, \quad Q = \sum_l y^{-l} B_l, \quad A_k, B_l \in L[u]$$

with u the residual variable. The bracket of weighted-homogeneous pieces is weighted-homogeneous, so $[P, Q] = \lambda x^2$ splits into one identity per weight:

$$\text{EIdent } A \, B \, \lambda \, n : \sum_{k+l=n} \text{LT}_{k,l}(A_k, B_l) = [n = 5] \cdot \lambda u^2, \quad \text{LT}_{k,l}(A, B) = kAB' - lA'B. \quad (4)$$

Layer k of P corresponds to $W = j - 2i$ via $n = 1 - W$; the raw equation key (i, j) corresponds to $(n = 2i + 1 - j, m = i)$, and the coefficient of the product term is $c = k \cdot (m + 1 - i_1) - l \cdot i_1$ with $l = n - k$ (§5.3 of the v3.1 guide). The top layer $n = 5$ carries the λu^2 term; all other layers are homogeneous.

4 Exact layer reduction

B0 (Lean, v1). From $\text{jac}(P, Q) = C \lambda \cdot x^2$ and NewtonNFC , the layer identities $\text{EIdent } A \, B \, \lambda \, n$ hold for every layer $n \geq -20$ (`layers_of_jac_c`, `layer_bracket_gen` in `LayersGen.lean`). This is an exact symbolic identity, verified by the Lean kernel.

B4 (Lean, v3). Coefficient extraction turns the polynomial identities $\text{EIdent } A \, B \, 0 \, n$, $n = 4, \dots, -2$, into scalar equations. For each monomial of $\text{LT}_{k,l}(A_k, B_l)$, `coeff_layerTermZ` computes its coefficient with integer indices; coefficients outside the branch-(c) supports vanish by `rangeP`, `rangeQ` (179 zero facts; $a_{0,0}$, $b_{0,0}$ are in the support but always carry coefficient 0). Each of the 132 raw equations then follows by `linear_combination`. This is the recipe of the repository's `BranchAbFinal.lean`, replayed by `certgen_c/gen_bridge_c.py` (`Descent2R/Bridge.lean`).

5 Normalization

B1 (Lean, v1). The top layer $n = 5$ is the branch-(a,b) problem, classified exactly over K_5 in [2] and formalized in the repository (`ChartProof/Final.lean`, `BranchAb*`). Branch (c) inherits its exact K_5 solution; this is where w and R enter (`eIdent_five`).

B2 (Lean, v1/v2). Degree-19 rigidity: $b_{12,24} \neq 0$ forces $t_2 = b_{12,22} \neq 0$. The theorems are `vertex_12_24_forces_t2` and `edge19_rigidity_of_jac` in `Edge19.lean`, and `t0_vertex_rigidity_of_jac` in `LayerE2.lean`, `Degree19.lean`.

B3 (Lean, v1). Torus and weighted scaling remove the continuous symmetries, normalizing to the rescaled K_5 top layer and the chart $b_{12,22} = 1$ (`eIdent_transport`, `main_theorem_c_of_claim` in `DescentClaim.lean`). These are `DescentClaimC`'s hypotheses: the 19 top-layer values, the 123 unknowns, and the 132 raw equations.

6 The top layer

At $n = 5$ the branch-(c) layer equations coincide with the branch-(a,b) top layer, whose complete K_5 classification is proved in [2] (five torus orbits; the degree-35 eliminant irre-

ducible over $\mathbb{Q}$). The classification is used here as a black box: the 19 top-layer values enter Descent2R/Main.lean as hypotheses h_v at minimal integer scale, and the K_5 arithmetic they satisfy is exactly the arithmetic of (2). No new top-layer analysis is needed for branch (c); the new mathematics is the descent of §7 and the chart analysis of §§8–10.

7 Descent

B5 (Lean, v3; kernel reflection). Each layer $n = 4, \dots, -2$ is a linear system in its new unknowns, with coefficients polynomial in the earlier unknowns and the chart parameters. The v2 operator-rank theorems (Rank/**, 96 modules) give each system's exact rank over K_5 ; the left null vectors give the elimination conditions:

Layer	Modules	Conditions produced
E_4 ($n = 4$)	Descent2R/L4 (54)	—
E_3 ($n = 3$)	L3 (57)	—
E_2 ($n = 2$)	L2c (57)	none (left null vector annihilates RHS)
E_1 ($n = 1$)	L1 (60)	Ω
E_0 ($n = 0$)	L0 (60)	Ψ and pure row E0pure_18_37
E_{-1} ($n = -1$)	Lm1 (99)	Φ_1, Φ_2 and pure rows Em1pure_17_36, Em1pure_18_38
E_{-2} ($n = -2$)	Lm2 (110)	$\Theta_1, \Theta_2, \Theta_3$ and pure rows Em2pure_16_35, Em2pure_17_37

lower_c.py computed these eliminations in Python; Descent2R replays them as 600 polynomial identities $\text{target} = \sum m_i \cdot \text{fact}_i$, each checked by the Lean kernel. Reflection uses the repository's Jacobian/ChartProof/Reflect.lean (built on core's Lean.Grind.CommRing): Expr/Poly, normalized by toPolyK and compared by decide +kernel. Each step theorem has the form $(\text{Expr.mul}(\cdot, \text{num } K) X). \text{denote } \text{ctx} = 0$ proved by lc_zero with a kernel decide; Main removes K with cancel_int when $K \neq 1$ (231 of 752 have). The context is 143 atoms in a balanced RArray (gctx2). A tactic proof of the same identities was tried first and abandoned: 214 MB of source, chunks over 10 min each. The kernel evaluates the normalizer directly with GMP arithmetic; decide +kernel is not native_decide—no compiler is trusted.

The main theorem of the descent (Descent2R/Main.lean):

Listing 2: chart_descent_refl: the twelve conditions from the 132 raw equations

```
-- Descent2R/Main.lean (statement; 752 have elided)
theorem chart_descent_refl
  (w : L) (hw : R w = 0)
  (v : Fin 19 → L) (h_v : v = K5_top_layer_values)
  (u : Fin 123 → L)
  (h4_ : raw_equations_n4) ... (hm2_ : raw_equations_nm2) :
  cond_Omega w ... = 0 \land cond_Psi w ... = 0 \land
  cond_Phi1 w ... = 0 \land cond_Phi2 w ... = 0 \land
  cond_Theta1 w ... = 0 \land cond_Theta2 w ... = 0 \land cond_Theta3 w ... = 0 \land
  cond_E0pure_18_37 w ... = 0 \land cond_Em1pure_17_36 w ... = 0 \land
  cond_Em1pure_18_38 w ... = 0 \land cond_Em2pure_16_35 w ... = 0 \land
  cond_Em2pure_17_37 w ... = 0
```

The twelve conjuncts are lower_c's conditions, defined once in CondsC.lean (generated by certgen_c/

gen_conds_c.py): each cond_* is the explicit polynomial with the same balanced sum tree as the reflected fact (subtrees of at most 32 terms are separate definitions: 110 definitions for 12 conditions), so ek.denote ctx = 0 and cond_* ... = 0 are definitionally equal. Bridge (Descent2R/Bridge.lean) then gives descentClaimC_of_chartEmpty: DescentClaimC $\Leftarrow$ ChartEmptyC.

8 The chart conditions

8.1 Chart variables

On the normalized chart $b_{12,22} = 1$ the conditions involve seven parameters:

Symbol	T_1	T_2	S_1	S_2	R_1	R_2	Q
Coefficient	$b_{11,20}$	$b_{12,22}$	$b_{11,21}$	$b_{12,23}$	$a_{6,13}$	$a_{7,15}$	$b_{10,21}$
Weight	1	1	2	2	3	3	4

8.2 The twelve conditions

certgen_c/conds_c.json (md5 168299d255af361578423c0ed4852c11, written by certgen_c/lower_c.py):

Condition	K_5 -terms	Integer terms (minimal scale, w expanded)
Ω	3	15
Ψ	32	160
Φ_1, Φ_2	54, 54	270, 270
$\Theta_1, \Theta_2, \Theta_3$	84, 84, 84	420, 420, 420
E0pure_18_37	9	41
Em1pure_17_36	39	195
Em1pure_18_38	12	60
Em2pure_16_35	77	381
Em2pure_17_37	55	275

8.3 The Ω parabola

Exactly over K_5 ,

$$\Omega = o_1 S_2^2 + o_2 T_2^2 S_2 + o_3 T_2^4, \quad o_2^2 = 4o_1 o_3, \quad o_1 \neq 0, \quad (5)$$

so that with $\kappa = -o_2/(2o_1) \in K_5$,

$$\Omega = o_1 (S_2 - \kappa T_2^2)^2. \quad (6)$$

The relation $o_2^2 = 4o_1 o_3$ is checked in step1/lowerc_chart.py and independently in step3c_chart_identity.py; $o_1(w) \neq 0$ is proved in the Lean kernel by a Bezout identity $a \cdot N_{o_1} + b \cdot R = c$ with c a 151-digit constant (T1Zero.s_X2). On the chart $T_2 = 1$, $\Omega = 0$ forces $S_2 = \kappa$: the chart collapses to the parabola's vertex line (Figure 3).

FIGURE PLACEHOLDER 3

What goes here (Figure 3). A schematic of the chart geometry near $\Omega = 0$. *Main panel:* the (T_2, S_2) -plane over K_5 (drawn schematically over $\mathbb{R}$ for illustration) showing the parabola $S_2 = \kappa T_2^2$ from (6), the chart line $T_2 = 1$ (normalization B3), and their intersection point $(1, \kappa)$ marked; the stratum $T_1 = b_{11,20} = 0$ is indicated as a transverse slice, with the two strata ($T_1 = 0$ handled in Lean, $T_1 \neq 0$ by the rank lemma) labelled. *Inset:* the six generators F_n on the chart ($T_2 = 1, S_2 = \kappa$) with their K_5 -term counts (22, 35, 35, 52, 52, 52) as a small bar chart, showing the input to the Macaulay matrix of §10. Rendering: vector PDF (TikZ), with a caption noting the schematic is over $\mathbb{R}$ for visualization while the mathematics is over K_5 .

9 The stratum $b_{11,20} = 0$

B7 (Lean kernel). On the stratum $T_1 = b_{11,20} = 0$ the conditions admit an exact K_5 certificate small enough for the kernel: degree 2, 75 K_5 -terms, coefficient heights at most 9,681 digits (chart_certificates/step1/certB_lowerc.json, produced by certB_lowerc.py via pivots from a mod-p RREF and an exact FLINT solve; independently rechecked by check_certB_lowerc.py with fmpq_mpoly expansion, w as a variable, reduced mod R at the end). T1Zero checks the certificate chain end to end in the kernel (gen_t1zero_lean.py):

Step	Identity (integer-polynomial, kernel-checked)	Build
s_sq	$D_0 D_{01} D_k^2 \Omega \equiv D_0 N_{01} X^2 \pmod{(T_2 - 1, R)}, X = D_k S_2 - N_k(w), D_k = 851565312$	1.5 s
s_X2	Bezout $a \cdot e_{sq} + b \cdot D_0 X^2 R = c \cdot D_0 X^2$ (c: 151 digits) $\Rightarrow X = 0$	—
s_f_n (6)	$D_k^d \cdot ck_n = D_k^d T_1 A_1 + D_k^d (T_2 - 1) A_2 + X A_3 + f_n + R A_4$	2.4–5.7 s
s_p_n (6)	$p_n = v_n f_n - q_n R$ (250–375 terms, $\approx 5,440$ digits)	17–28 s
s_final	$\sum_n p_n = DD$, a 5387-digit integer, exact	55.6 s

chartEmpty_t1_zero chains these: $(DD : L) = 0$ contradicts characteristic 0 (26 s under lake). The hypotheses are exactly the `cond_*` of `CondsC`, definitionally the conditions `Descent2R` derives. Five perturbation controls (Bezout constant +1, a dropped equation, a perturbed cofactor coefficient, $DD + 1$, a dropped T_2 equation) are all rejected by the kernel (`logs/t1z_controls.log`: five of five “Tactic decide failed”, as required).

B6 (Lean). `chartEmptyC_of_T1ne0 (T1Zero/Combine.lean)` splits on $b_{11,20}$: `ChartEmptyC` $\Leftarrow$ `ChartEmptyC_T1ne0`, isolating exactly what Lean does not check as a precise `Prop`:

Listing 3: ChartEmptyC_T1ne0: what Lean labels, proved in §10

```
-- T1Zero/Combine.lean (exact statement)
def ChartEmptyC_T1ne0 (L : Type*) [Field L] [CharZero L] : Prop :=
  \forall (w : L), R w = 0 \to b_12_22 = 1 \to b_11_20 \ne 0 \to
    \not (cond_Omega w ... = 0 \land ... \land cond_Theta3 w ... = 0)
```

10 The whole chart

A canonical K_5 certificate for the whole chart is too large to build on this machine: at weight $W = 24$ the Hadamard bound is 4,392,160 digits, and calibrating against the $t_1 = 0$ slices ($W = 2, 3, 4$, where the true-to-bound height ratios are 0.061, 0.061, 0.046) gives heights of $2.0\text{--}2.7 \times 10^5$ digits, about 2×10^9 digits in all (grade C estimate; the Hadamard bound itself is rigorous). Exact solution runs out of memory at 7 GB; multimodular reconstruction would need $\approx 3 \times 10^4$ primes at 15–20 s each, about six days. Instead we prove the *existence* of the certificate.

10.1 The rank lemma

Lemma 10.1 (Fixed-degree Macaulay rank lifts). *Let $A = \mathbb{Z}_{(p)}[w]/(R)$, free of rank 5 over $\mathbb{Z}_{(p)}$, embedding in K_5 . Let $\varphi : A \rightarrow \mathbb{F}_p$, $w \mapsto w_0$, with $R(w_0) \equiv 0 \pmod{p}$. Let M be the weighted Macaulay matrix of the chart generators at fixed weight W (columns (n, m) with $\text{wdeg } m \leq W - \text{wt } F_n$; rows the monomials of the products $m \cdot F_n$, plus 1; entries the coefficients). If $\varphi(M)$ has full row rank N , then M has full row rank over K_5 .*

Proof. Some $N \times N$ minor satisfies $\det \varphi(M_S) = \varphi(\det M_S) \neq 0$. Hence $\det M_S \neq 0$ in $A \subset K_5$, and M_S is invertible over the field K_5 . $\square$ $\square$

Consequences. $Mx = e_1$ is solvable over K_5 , so $1 = \sum c_n F_n$ in $K_5[t, s, r, u, q]$ with $\text{wdeg } c_n \leq 24 - \text{wt } F_n$. Mapping $K_5 \rightarrow L$ by $w \mapsto w$, no L admits a common zero. With $S_2 = \kappa$ from (6) ($\alpha_1 \neq 0$), no chart point satisfies $\Omega = \Psi = \dots = \Theta_3 = 0$: that is `ChartEmptyC`, hence `ChartEmptyC_T1ne0`, even without the pure rows.

Remark 10.2 (Why this lemma succeeds where the retracted one failed). A unit ideal mod p says only that the Macaulay system is *consistent* mod p ; consistency does not lift ($1 + 109X$ over $\mathbb{Z}_{(109)}$). Full row rank at fixed degree is the nonvanishing of a minor, an *open* condition, and open conditions lift. The planted-zero control below shows the sensitivity: one common zero costs exactly one rank.

10.2 The instance

`step3b_rank_lift.py` (6.5 min, 0.5 GB; `step3b_rank_lift.log`):

	$p = 1000003$ $w_0 = 806739$	$p = 32003$ $w_0 = 11147$
All coefficients p -integral, $R(w_0) \equiv 0$	asserted	asserted
$W = 22$: rank / augmented / rows	2277/2278/2281: inconsistent	same
$W = 23$: rank / augmented / rows	2707/2708/2708: inconsistent	same
$W = 24$: 3199×6054	3199/3199/3199: full row rank	same
NumPy pivot-block elimination (independent)	$\det \neq 0$	$\det \neq 0$
Planted common zero, $W = 24$ (control)	rank 3198, inconsistent	same

The protocol was frozen before the run: the lemma, the second prime, the independent NumPy elimination and the controls were all fixed in `step3b_rank_lift.py` before it executed. The $W = 24$ threshold is sharp: $W = 22$ and $W = 23$ are inconsistent (Figure 4).

FIGURE PLACEHOLDER 4

What goes here (Figure 4). A grouped bar chart of the Macaulay rank profile at weights $W = 22, 23, 24$, for both primes ($p = 1000003$ and $p = 32003$). For each weight: bars for the number of rows (2281, 2708, 3199), the computed rank (2277, 2707, 3199), and the augmented rank, with the $W = 24$ group highlighted (full row rank 3199/3199, the obstruction). A fourth group shows the planted-common-zero control at $W = 24$ (rank 3198, exactly one short). Data source: `step3b_rank_lift.log`, reproduced from `chart_certificates/step3b_rank_lift.py`. Rendering: vector PDF from a short Python script reading the log's PASS lines, so the figure regenerates from the executed output; the y-axis starts at a truncated range with a break mark to make the single-rank deficit visible.

The chart generators are independently verified as the right polynomials (exact evaluation at 12 random points; an independent κ ; the exact Ω square; controls including $\kappa + 1$ detected) by the script `step3c_chart_identity.py`. This is **B8**, proved outside Lean at grade B.

11 Proof of the main theorem

Listing 4: main_theorem_c_of_chartEmpty_T1ne0

```
-- Jacobian/BranchC/Combine (exact statement)
theorem main_theorem_c_of_chartEmpty_T1ne0
  {L : Type*} [Field L] [CharZero L]
  (h : ChartEmptyC_T1ne0 L) :
  \not \exists (P Q : MvPolynomial (Fin 2) L) (lam : L),
    lam \ne 0 \land NewtonNFC P Q \land jac P Q = C lam * X 0 ^ 2
```

Proof. Let P, Q, λ satisfy `NewtonNFC` and $[P, Q] = \lambda x^2$, $\lambda \neq 0$. B0–B3 give `DescentClaimC`'s hypotheses (the 19 top-layer values, 123 unknowns, 132 raw equations). B4 extracts the raw equations; B5 (`chart_descent_refl`) yields the twelve conditions, i.e. `ChartEmptyC`. B6 splits on $b_{11,20}$: on $b_{11,20} = 0$, B7 (`chartEmpty_t1_zero`) gives `False` in the kernel; on $b_{11,20} \neq 0$, the hypothesis `ChartEmptyC_T1ne0` gives `False`. Hence no such P, Q, λ exist. $\square$

Corollary 1.3 follows by supplying `ChartEmptyC_T1ne0` from §10; Corollary 1.4 by applying [1, Proposition 4.3].

The bridge composition is summarized in Table 11.

Bridge	From	To	Lean name	Grade
B0	$[P, Q] = \lambda x^2$, NewtonNFC	layer identities E_n	layers_of_jac_c	A
B1	E_5 , vertices	K_5 top layer	eIdent_five	A
B2	B0, B1	$t_2 = b_{12,22} \neq 0$	vertex_12_24_forces_t2	A
B3	B0–B2	chart $b_{12,22} = 1$	main_theorem_c_of_claim	A
B4	E_n , $n = 4 \dots -2$	132 raw equations	eIdent_c4... eIdent_cm2	A
B5	132 equations	twelve conditions	chart_descent_refl	A
B6	ChartEmptyC_T1ne0	ChartEmptyC	chartEmptyC_of_T1ne0	A
B7	conditions, $b_{11,20} = 0$	False	chartEmpty_t1_zero	A
B8	chart generators	False	rank lemma + step3b_rank_lift.py	B

12 Epistemic status and what is not claimed

HLRE bookkeeping: grades A (machine-checked or exact proof from stated axioms), B (derived under explicit premises), C (heuristic or fitted), D (tautological or refuted).

ID	Claim	Evidence	Grade
C1	main_theorem_c_of_claim	Lean kernel (v1)	A
C2	descentClaimC_of_chartEmpty	Lean kernel, reflective; 600 identities pre-checked in Python	A
C3	chartEmpty_t1_zero	Lean kernel; FLINT certificate rechecked in Python; 5 controls	A
C4	chartEmptyC_of_T1ne0	Lean	A
C5	ChartEmptyC holds (all char.-0 L)	rank lemma + two-prime, two-implementation ranks + input checks	B
C6	canonical certificate needs $\approx 2 \times 10^9$ digits	Hadamard bound (rigorous) $\times$ calibrated ratio (fitted)	C
C7	the stated reduction lemma is inconsistent	Lean counterexample (1 + 109X)	A
C8	lower_c's ideal differs from the pipeline's	exact comparison	A

Verbs follow HLRE control: “proves” only for Lean-checked or exact deductions; “proves under stated computational premises” for C5; “estimates” for C6.

What is *not* claimed: not the Jacobian conjecture; not any other case of GGHV Proposition 4.3; not the reduction from the conjecture to Proposition 4.3; not a Lean proof of the stratum $b_{11,20} \neq 0$; not an explicit K_5 certificate for the whole chart (only its existence, C5); not a review of the deferred single-prime p-adic argument or of the retracted char0_cert bundle. The mod-p Gröbner/lift certificates ($\mathbb{F}_{101}$, $\mathbb{F}_{1000003}$, $\mathbb{F}_{1095}$) are certificates, not proofs: they exclude only solutions integral at that prime. The rank argument is what lifts.

Open items, in order: (1) I3 replication of C5—an independent rank computation of the $W = 24$ Macaulay matrix in Magma, Sage or Singular from conds_c.json and κ ; (2) optional formalization of the rank lemma's instance in Lean (RingHom.map_det; the instance needs the 3199×6054 matrix as Lean data); (3) deferred reviews; (4) the v2 corrected-commit proposal for branch_c/, on the author's explicit word only.

13 Conclusion

Branch (c), the last open configuration of GGHV Proposition 4.3 at degree pair (72, 108), is eliminated. The descent from $[P, Q] = \lambda x^2$ to twelve explicit chart conditions is proved twice: symbolically, as 600 kernel-checked reflective identities in Lean 4 over 505 generated modules, and numerically, by exact finite-field linear algebra. The chart is empty: in the Lean kernel on the stratum $b_{11,20} = 0$, and by the rank lemma—full row rank of the $W = 24$ Macaulay matrix at two primes, verified by two independent implementations—on the whole chart. An earlier, false reduction lemma is retracted and replaced by the rank lemma; the paper’s logical dependencies are exactly those of Table 12. With branches (a,b) eliminated in [2], no (72, 108) counterexample arises from GGHV Proposition 4.3, conditional on that proposition’s correctness and exhaustiveness.

14 Prior work, attribution, and independence

The GGHV classification [1] is the sole external mathematical input, used only in Corollary 1.4. The top-layer K_5 classification and the reflection library are inherited from the branch-(a,b) formalization [2] without modification; `verify_branch_c_lean.sh` step 0 checks these prerequisites by content fingerprint. The `branch_c/` pipeline scripts were read-only throughout: the pipeline parametrizes the chart differently, and the v3 proof uses only `lower_c`’s conditions, derived in Lean. The $W = 24$ rank at $p = 1000003$ was known from the step-3 measurement; the lemma, the second prime (32003), the independent NumPy elimination and the controls were frozen in `step3b_rank_lift.py` before it ran—the second prime agreed, as expected, since full row rank over K_5 implies full row rank at all but finitely many primes.

References

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- [9] H. Helali, *Exact characteristic-zero certificate for the GGHV (72, 108) configuration*, Zenodo doi:10.5281/zenodo.21479814 (2026).
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A Computational archive

All objects cited in this paper are archived in the public repository [2D-Jacobian-Conjecture-Workspace](#), directory `JC2D-branches_a_b_c_unified/branch_c/`:

- `bundle_v3_1/` — the v3.1 bundle, byte-for-byte: the Lean overlay (`jacobian_lean/`: `Jacobian/BranchC/**`, `certgen_c/`, `chart_certificates/`, `verify_branch_c_lean.sh`, `BRANCH_C_LEAN_STATUS.md`), every log (`logs/`), diagnostics (`diag/`), and the v2 carryover;
- `chart_certificates/step1/` — the exact K_5 certificate for $t_1 = 0$ (`certB_lower_c.json`) with its independent checker;
- `chart_certificates/step3b_rank_lift.py` — the frozen two-prime, two-implementation rank computation (§10);
- `chart_certificates/step3c_chart_identity.py` — the independent chart-generator checks;
- `certgen_c/` — the generators (`gen_refl_c.py`, `gen_bridge_c.py`, `gen_conds_c.py`, `gen_t1zero_lean.py`) with byte-identical regeneration;
- `muse_refutation/RefuteReductionLemma.lean` — the Lean derivation of False from the retracted reduction-lemma axiom (Remark 1.6).

The v3.1 guide `BRANCH_C_GUIDE.md` is the complete 13-section reference for the bundle, including the log index (§9), diagnostics (§10), open items (§11) and reproduction instructions (§12).

Reproducibility

Lean: overlay the `jacobian_lean/` directory on the repository’s branch-(a,b) Lean project (Lean 4.34.0, Mathlib v4.34.0), then `./verify_branch_c_lean.sh` (exit 0: prerequisite fingerprints, byte-identical regeneration, all 790 modules up to date, axioms audit, no sorry/axiom/native_decide). Outside Lean (python3 with python-flint, NumPy): `chart_certificates/step1/run_step1.sh`, `python3step3c_chart_identity.py`, `python3step3b_rank_lift.py` (6.5 min, 0.5 GB). `#print axioms` on the 53 audited theorems reports only `propext`, `Classical.choice`, `Quot.sound` (`jacobian_lean/axioms_branch_c.log`).

B Lean module map

Group	Modules	Key theorems
v1/v2 branch-(c)	162	layers_of_jac_c, eIdent_five, vertex_12_24_forces_t2
Descent2R	505 + Main	chart_descent_refl (752 haves)
CondsC, Bridge	2	cond_* (110 definitions), descent-ClaimC_of_chartEmpty
T1Zero	14 + Main	chartEmpty_t1_zero
Combine	1	chartEmptyC_of_T1ne0, main_theorem_c_of_chartEmpty_T1ne0

Build: Descent2R 2.9 h of module time (peak 5.6 GB RSS); Main 2,298 s; Bridge 269 s; CondsC 175 s; T1Zero.Main 26 s. Verified by `verify_branch_c_lean.sh`, exit 0 (`logs/verify_v3.log`).

C Computational and AI-assisted research disclosure

Muse (Meta), the author’s personal AI assistant, assembled the v3.1 bundle, wrote the generators and the verification scripts, drafted this manuscript, and discovered the counterexample refuting the stated reduction lemma; all computational outputs were verified by execution, by independent reimplementations, or by the Lean kernel. An external reviewer assessed the T1Zero certificate chain and the rank lemma as “absolutely flawless” in their respective domains, and identified the rank computation’s non-Lean status as the one point a hostile referee can press (I3 replication proposed in §12). The author (B. D. McCrary) directed the investigation, verified the results, and takes responsibility for the manuscript. The mathematical argument stands on the artifacts above, not on these attributions.